# nRF9151 Connect Kit


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## Revision History

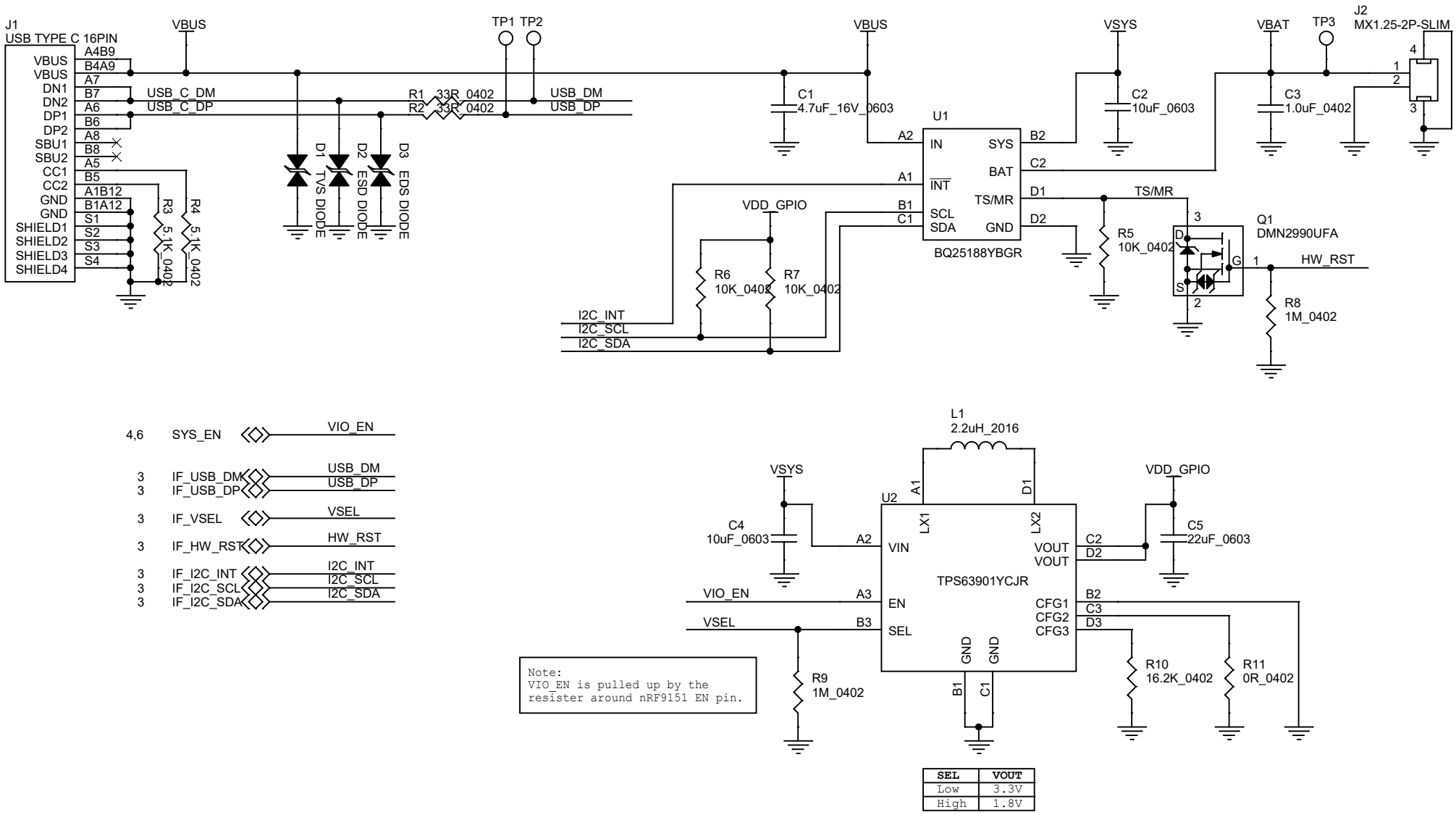
| Rev. | Changes | Date |
|---|---|---|
| Rev A | The initial public release | Dec-25-2024 |
| Rev A1 | Updated to nRF9151 A1 Module with NTN support<br><br>No changes to PCB | Jun-18-2026 |
| Rev A2 | Updated to BQ25188 with better performance and material availability<br><br>No changes to PCB | Nov-10-2026 |


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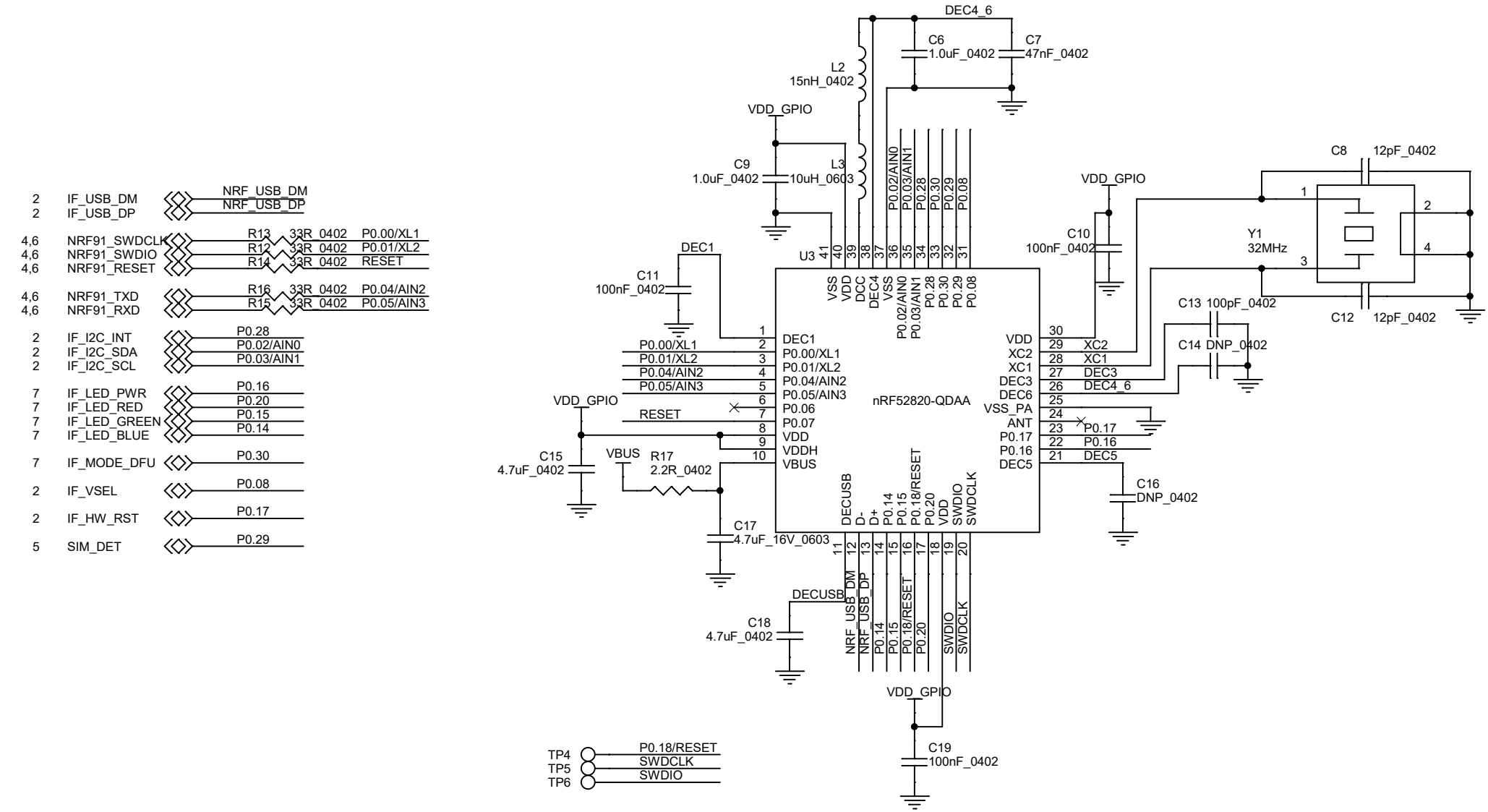

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|---|---|---|
| Title: Title Sheet | | |
| Size: A4 | Document Number: NCK9151SCH | Rev: A |
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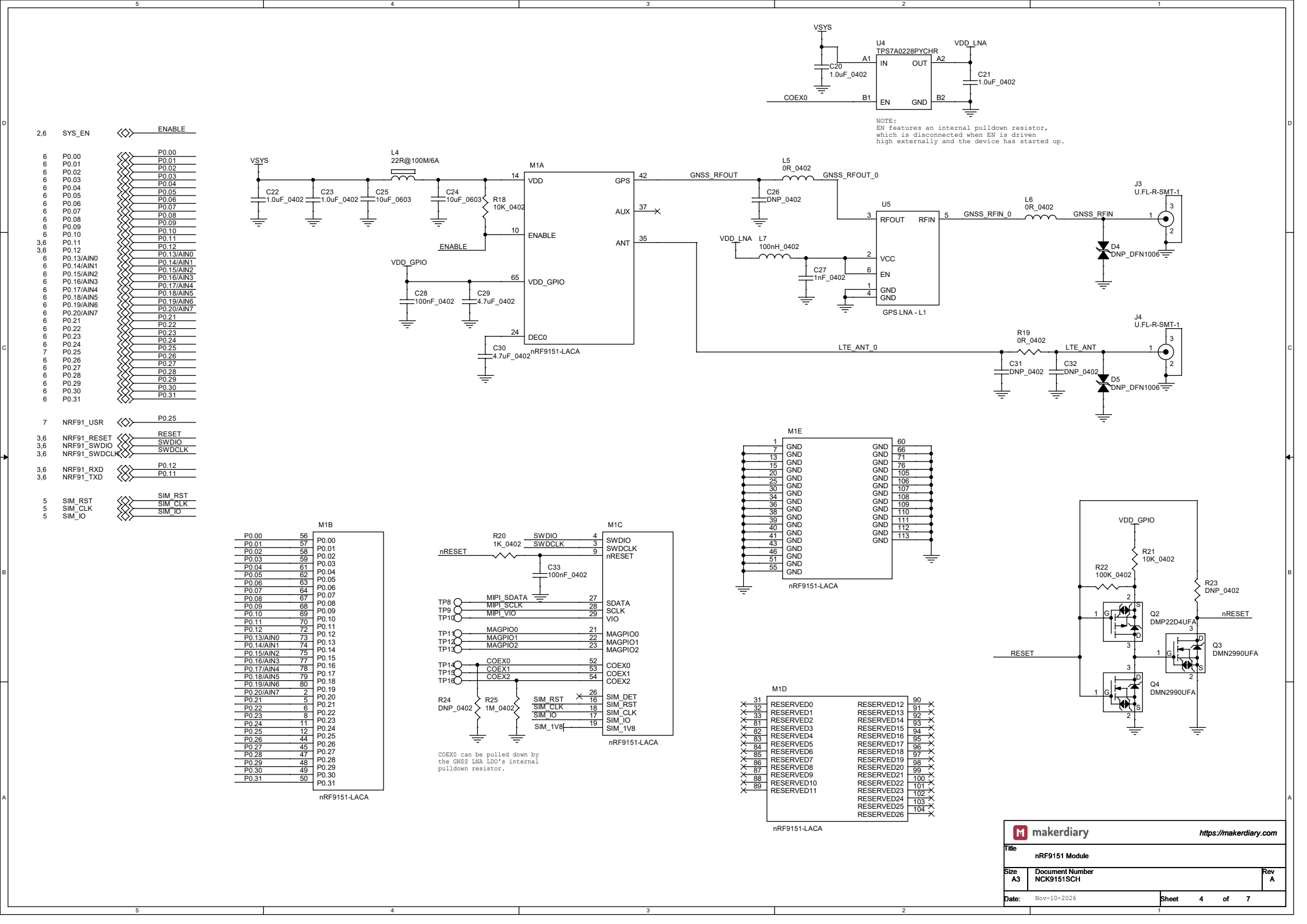

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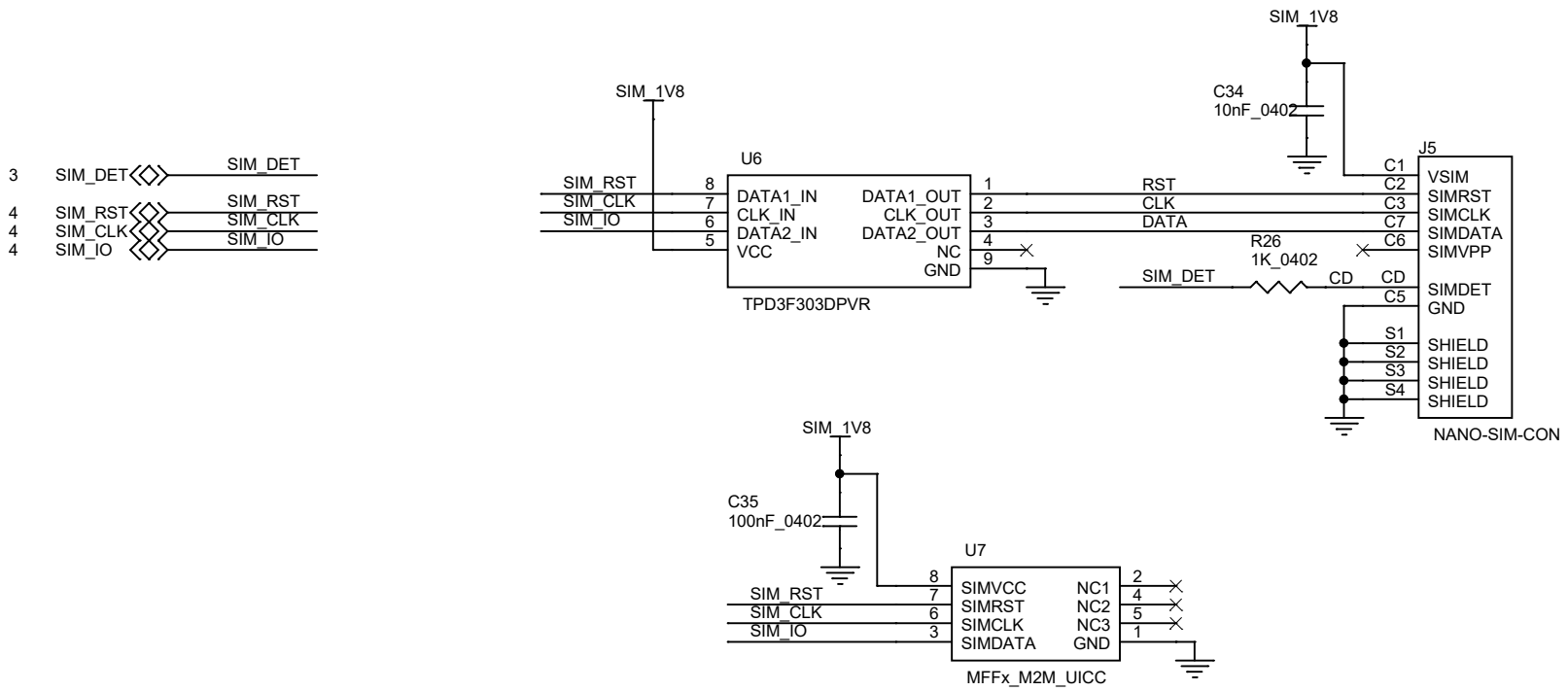
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|---|---|---|
| Title | Power Supply | |
| Size A4 | Document Number NCK9151SCH | Rev A |
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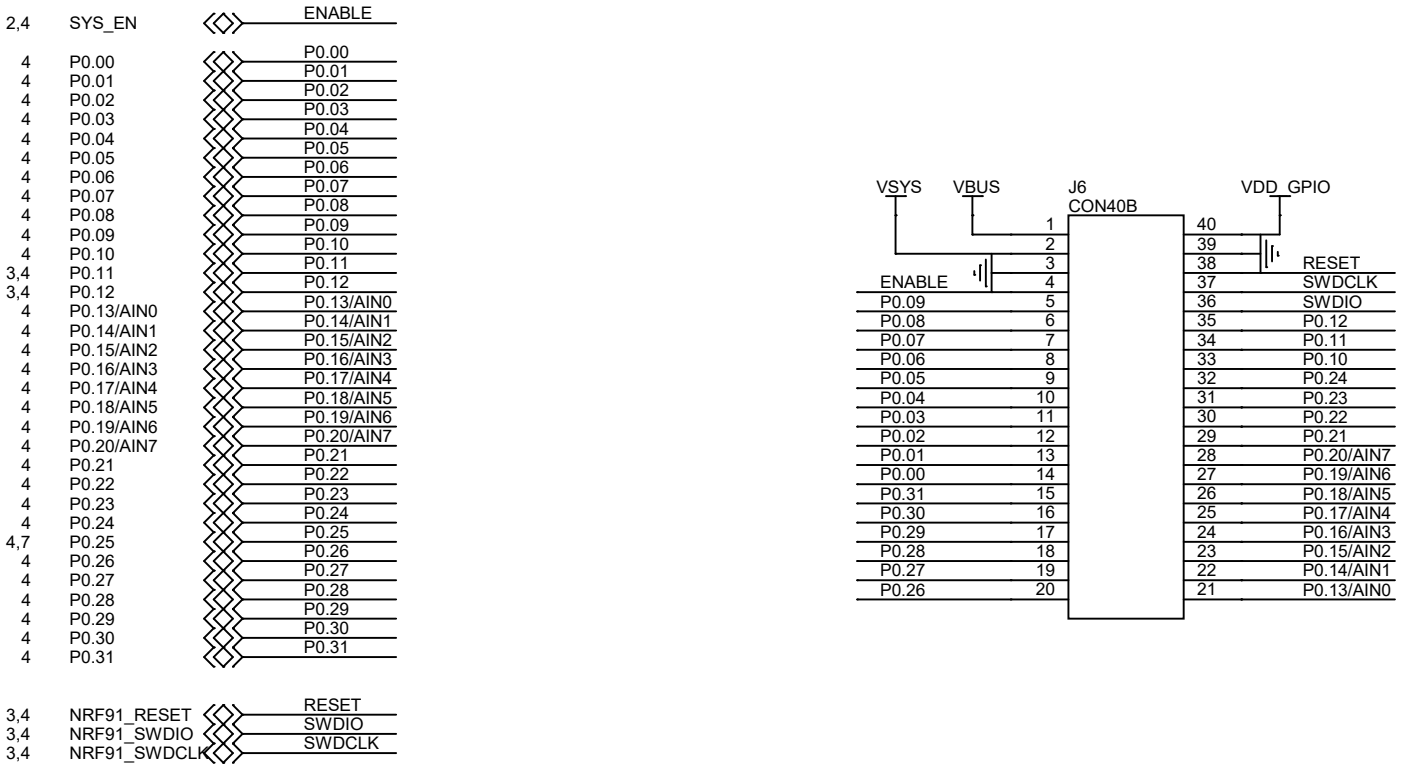
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|---|---|---|
| Title | Interface MCU - NRF52820 | |
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2,6 SYS_EN ENABLE
6 P0.00 P0.00
6 P0.01 P0.01
6 P0.02 P0.02
6 P0.03 P0.03
6 P0.04 P0.04
6 P0.05 P0.05
6 P0.06 P0.06
6 P0.07 P0.07
6 P0.08 P0.08
6 P0.09 P0.09
6 P0.10 P0.10
3,6 P0.11 P0.11
3,6 P0.12 P0.12
6 P0.13/AIN0 P0.13/AIN0
6 P0.14/AIN1 P0.14/AIN1
6 P0.15/AIN2 P0.15/AIN2
6 P0.16/AIN3 P0.16/AIN3
6 P0.17/AIN4 P0.17/AIN4
6 P0.18/AIN5 P0.18/AIN5
6 P0.19/AIN6 P0.19/AIN6
6 P0.20/AIN7 P0.20/AIN7
6 P0.21 P0.21
6 P0.22 P0.22
6 P0.23 P0.23
6 P0.24 P0.24
6 P0.25 P0.25
7 NRF91_USR P0.25
6 P0.26 P0.26
6 P0.27 P0.27
6 P0.28 P0.28
6 P0.29 P0.29
6 P0.30 P0.30
6 P0.31 P0.31
3,6 NRF91_RESET RESET
3,6 NRF91_SWDIO SWDIO
3,6 NRF91_SWDCLK SWDCLK
3,6 NRF91_RXD P0.12
3,6 NRF91_TXD P0.11
5 SIM_RST SIM_RST
5 SIM_CLK SIM_CLK
5 SIM_IO SIM_IO
VSYS
U4
TPS7A0228PYCHR
IN OUT EN GND
A1 A2 B1 B2
C20 1.0uF_0402
C21 1.0uF_0402
VDD_LNA
COEX0
NOTE:
EN features an internal pulldown resistor,
which is disconnected when EN is driven
high externally and the device has started up.
L4
22R@100M/6A
C22 1.0uF_0402
C23 1.0uF_0402
C25 10uF_0603
C24 10uF_0603
R18 10K_0402
M1A
VDD GPS AUX ANT DEC0 ENABLE VDD_GPIO
14 42 37 35 24 10 65
GNSS_RFOUT
L5 0R_0402
GNSS_RFOUT_0
C26 DNP_0402
U5
RFOUT RFIN VCC EN GND GND
3 5 2 6 1 4
GPS LNA - L1
L6 0R_0402
GNSS_RFIN_0
GNSS_RFIN
J3
U.FL-R-SMT-1
D4 DNP_DFN1006
VDD_LNA
L7 100nH_0402
C27 1nF_0402
VDD_GPIO
C28 100nF_0402
C29 4.7uF_0402
C30 4.7uF_0402
nRF9151-LACA
LTE_ANT_0
R19 0R_0402
LTE_ANT
C31 DNP_0402
C32 DNP_0402
D5 DNP_DFN1006
J4
U.FL-R-SMT-1
M1B
P0.00 56 P0.01 57 P0.02 58 P0.03 59 P0.04 61 P0.05 62 P0.06 63 P0.07 64 P0.08 67 P0.09 68 P0.10 69 P0.11 70 P0.12 72 P0.13/AIN0 73 P0.14/AIN1 74 P0.15/AIN2 75 P0.16/AIN3 77 P0.17/AIN4 78 P0.18/AIN5 79 P0.19/AIN6 80 P0.20/AIN7 2 P0.21 5 P0.22 6 P0.23 8 P0.24 11 P0.25 12 P0.26 44 P0.27 45 P0.28 47 P0.29 48 P0.30 49 P0.31 50
nRF9151-LACA
M1C
R20 1K_0402
SWDIO 4
SWDCLK 3
nRESET 9
C33 100nF_0402
TP8 TP9 TP10 MIPI_SDATA 27 MIPI_SCLK 28 MIPI_VIO 29 SDATA SCLK VIO
TP11 TP12 TP13 MAGPIO0 21 MAGPIO1 22 MAGPIO2 23 MAGPIO0 MAGPIO1 MAGPIO2
TP14 TP15 TP16 COEX0 52 COEX1 53 COEX2 54 COEX0 COEX1 COEX2
R24 DNP_0402
R25 1M_0402
SIM_RST 26 16 SIM_DET SIM_RST
SIM_CLK 18 SIM_CLK
SIM_IO 17 SIM_IO
SIM_1V8 19 SIM_1V8
nRF9151-LACA
COEX0 can be pulled down by
the GNSS LNA LDO's internal
pulldown resistor.
M1E
GND 1 7 13 15 20 25 30 34 36 38 39 40 41 43 46 51 55
GND 60 66 71 76 105 106 107 108 109 110 111 112 113
nRF9151-LACA
M1D
RESERVED0 31 RESERVED1 32 RESERVED2 33 RESERVED3 81 RESERVED4 82 RESERVED5 83 RESERVED6 84 RESERVED7 85 RESERVED8 86 RESERVED9 87 RESERVED10 88 RESERVED11 89
RESERVED12 90 RESERVED13 91 RESERVED14 92 RESERVED15 93 RESERVED16 94 RESERVED17 95 RESERVED18 96 RESERVED19 97 RESERVED20 98 RESERVED21 99 RESERVED22 100 RESERVED23 101 RESERVED24 102 RESERVED25 103 RESERVED26 104
nRF9151-LACA
VDD_GPIO
R21 10K_0402
R22 100K_0402
R23 DNP_0402
Q2 DMP22D4UFA
Q3 DMN2990UFA
Q4 DMN2990UFA
nRESET
RESET
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Title
nRF9151 Module
Size A3
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| | | |
|---|---|---|
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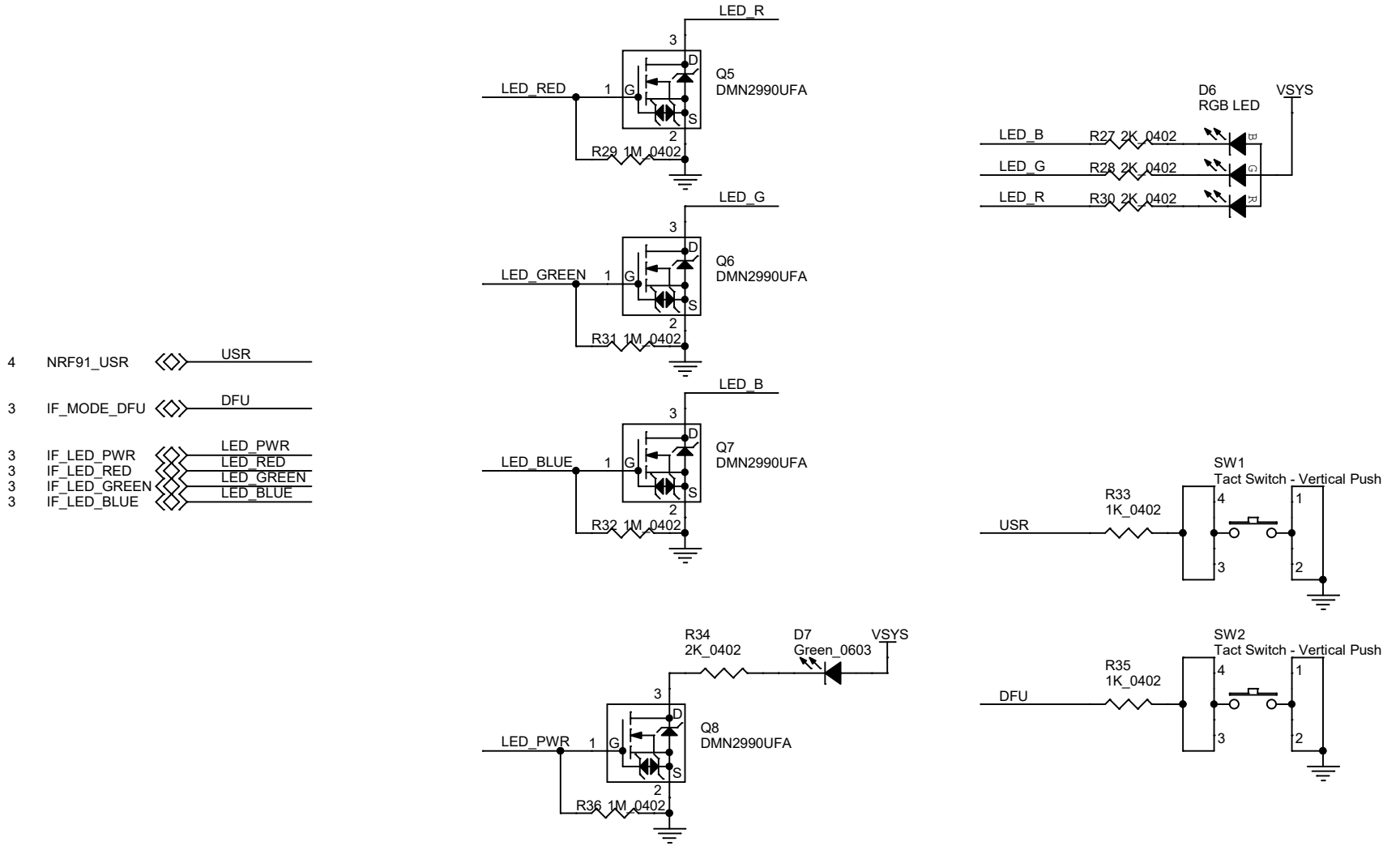






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| | | |
|---|---|---|
| Title: Pinout | | |
| Size: A4 | Document Number: NCK9151SCH | Rev: A |
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|---|---|---|
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| Title | LEDs and Buttons | |
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